



Taxi Driver, 1976, Martin Scorsese.



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Official Course Outline

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Course Outline (Syllabus)

3. Solution Design and Implementation



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3. Solution Design & Implementation ▾

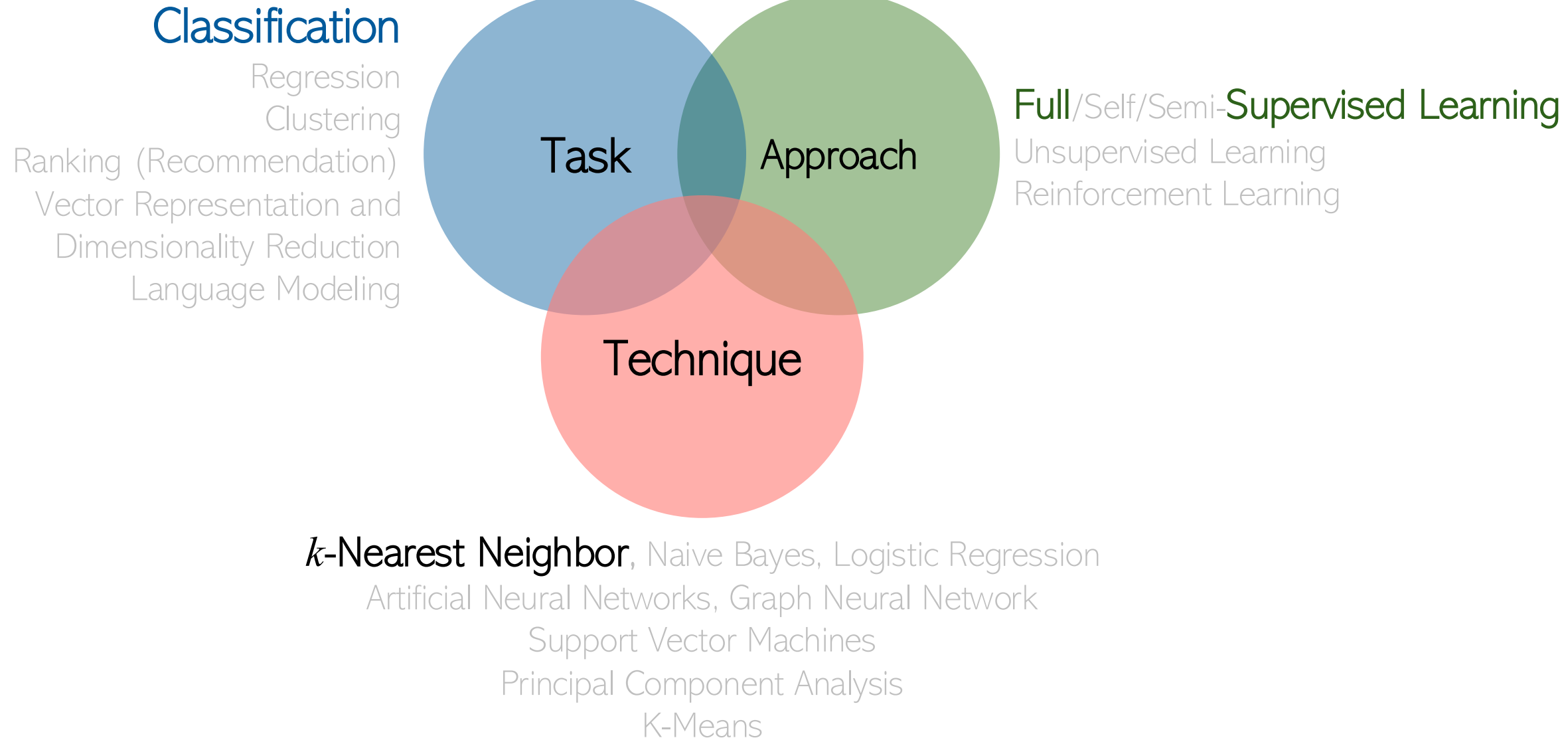


Assignment

Due November 5 at 6:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

4 Outline: W02: Topics Start



Outline: W05B

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Embeddings ...
- III) Evaluation

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 k -Nearest Neighbor: Word-Document Matrix

Word Space $\rightarrow$ Document Space $\rightarrow \mathbb{R}^{n=|\mathbf{D}|}$

	As You Like It	Twelfth Night	Julius Caesar	Henry V	
battle	1	0	7	13	Word Point
good	114	80	62	89	
fool	36	58	1	4	
wit	20	15	2	3	
					$ \mathbf{V} \times \mathbf{D} $

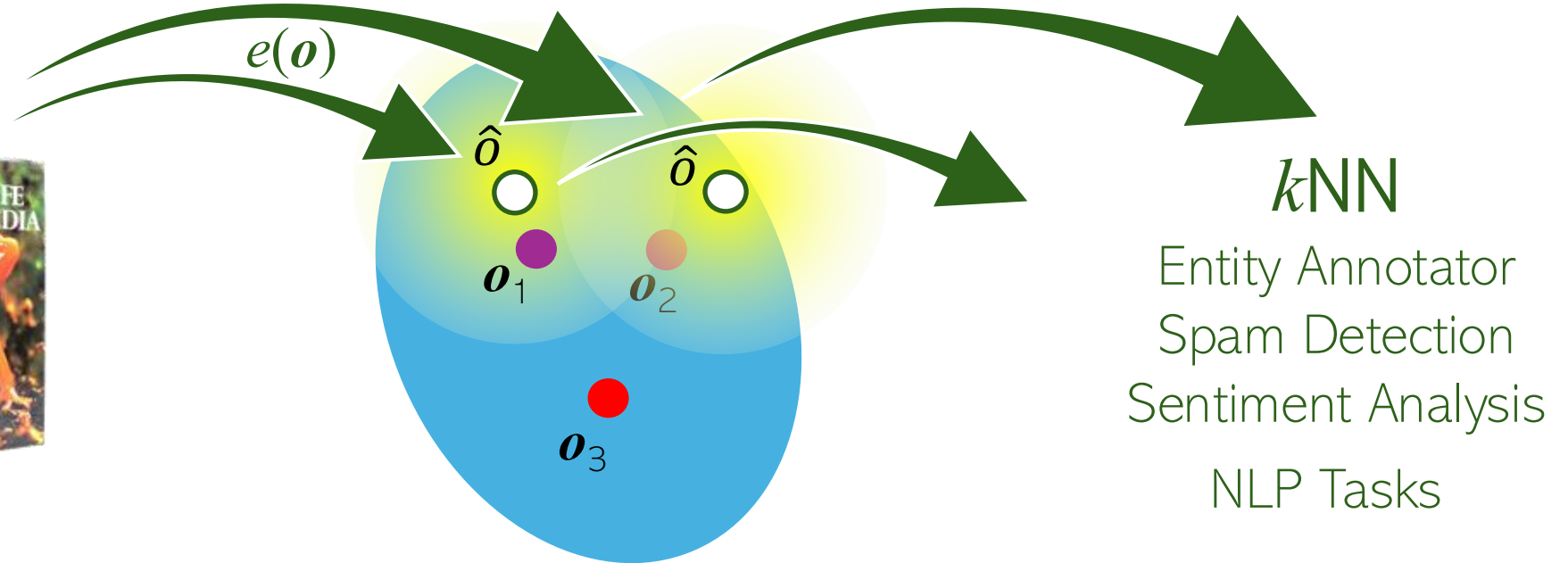
Doc Point

$\mathbb{R}^{n=|\mathbf{V}|} \leftarrow$ Word Space $\leftarrow$ Document Space

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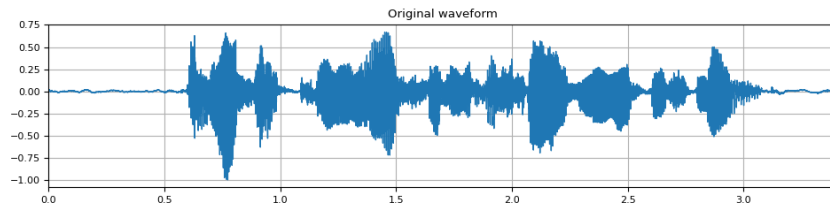
k -Nearest Neighbor: Transfer Learning

Can we have *multiple* points for a word?

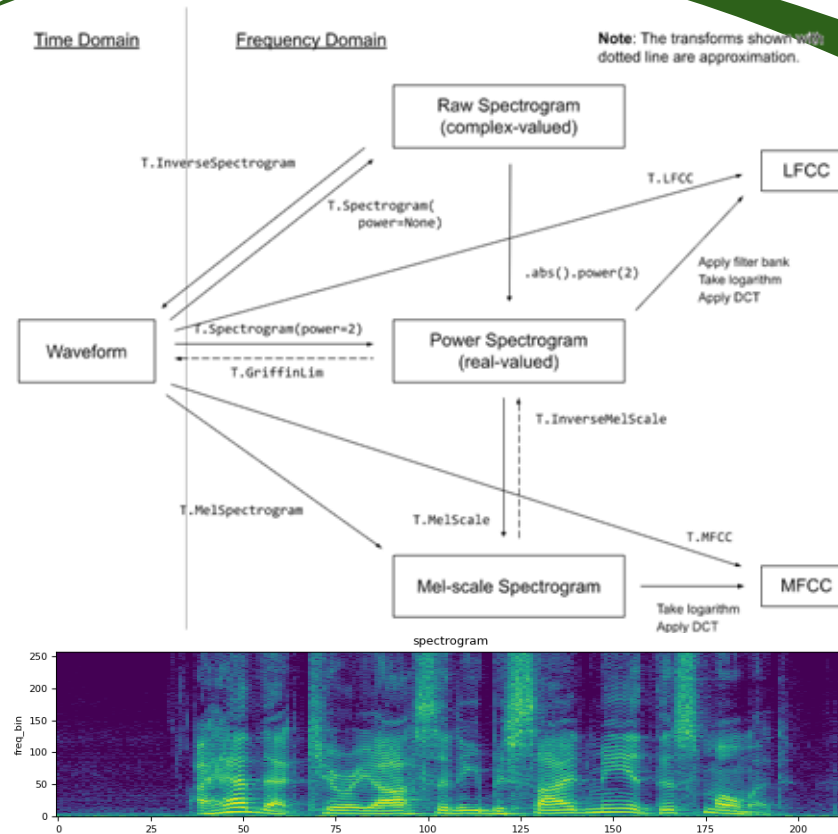
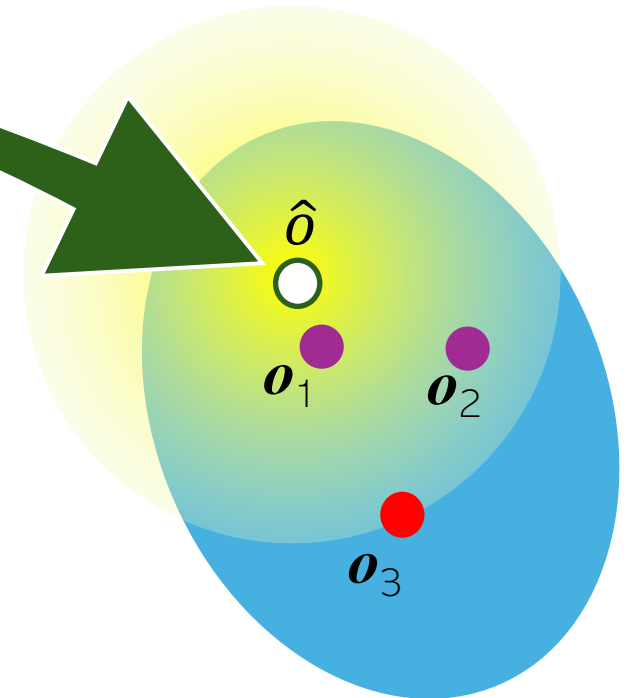


Apple (fruit) and Apple (company)
Python (snake) and Python (programming language)
Zyzzzyva (fly) and Zyzzzyva (magazine)
Placebo (dummy drug) and Placebo (music band)

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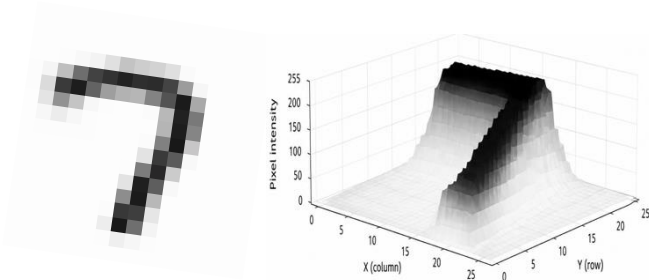
 k -Nearest Neighbor: Audio (Voice, Music, ...)

Wave in Time Space
 $\mathbb{R}^{n=|\text{Time}|}$

 $e(o)$ 

Fast Fourier Transform (FFT) in Frequency Space
 $\mathbb{R}^{n=|\text{FFT Coefficient}|}$

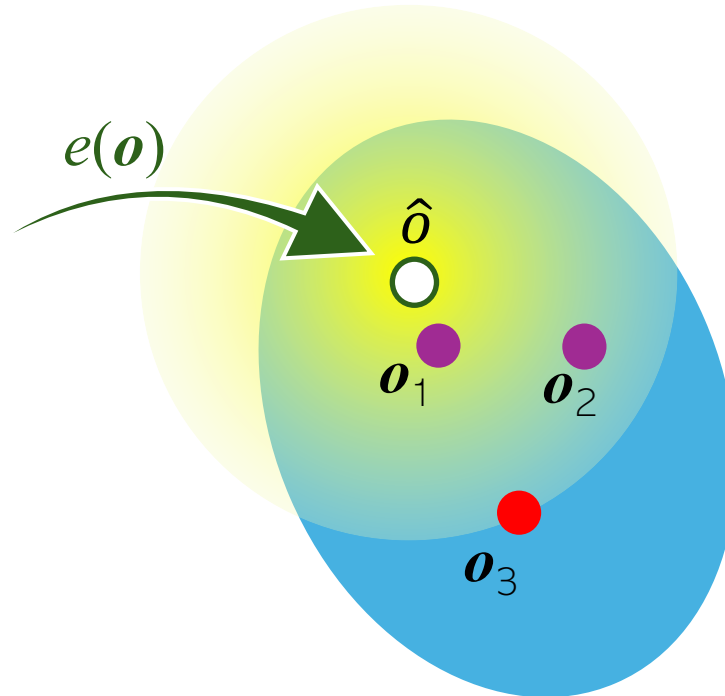
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 k -Nearest Neighbor: Image (MNIST Digits)

[	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	80	220	255	255	255	255	255	255	220	80	0	0	0]
[	0	0	0	0	0	0	0	0	90	255	255	0	0	0]
[	0	0	0	0	0	0	0	0	0	210	210	0	0	0]
[	0	0	0	0	0	0	0	0	140	255	90	0	0	0]
[	0	0	0	0	0	0	0	90	255	180	0	0	0	0]
[	0	0	0	0	0	0	80	255	200	0	0	0	0	0]
[	0	0	0	0	0	50	240	240	0	0	0	0	0	0]
[	0	0	0	0	0	180	255	90	0	0	0	0	0	0]
[	0	0	0	0	0	255	140	0	0	0	0	0	0	0]
[	0	0	0	0	140	255	0	0	0	0	0	0	0	0]
[	0	0	0	0	220	180	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0]

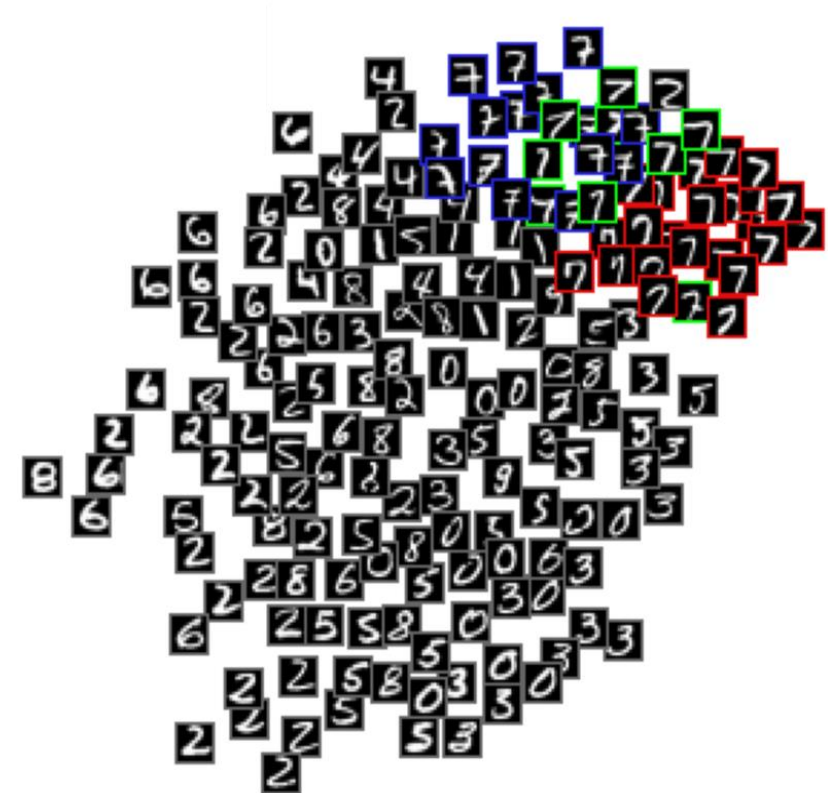
2D Wave in Grayscale Pixel Space

$$\mathbb{R}^{n=|\mathbf{Height}|} \times \mathbb{R}^{n=|\mathbf{Width}|}$$



Flatten row/column wise

$$\mathbb{R}^{n=|\mathbf{Height}| \times |\mathbf{Width}|}$$

*"What if we have colored pixel space?"*

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k -Nearest Neighbor: Image (MNIST Digits)

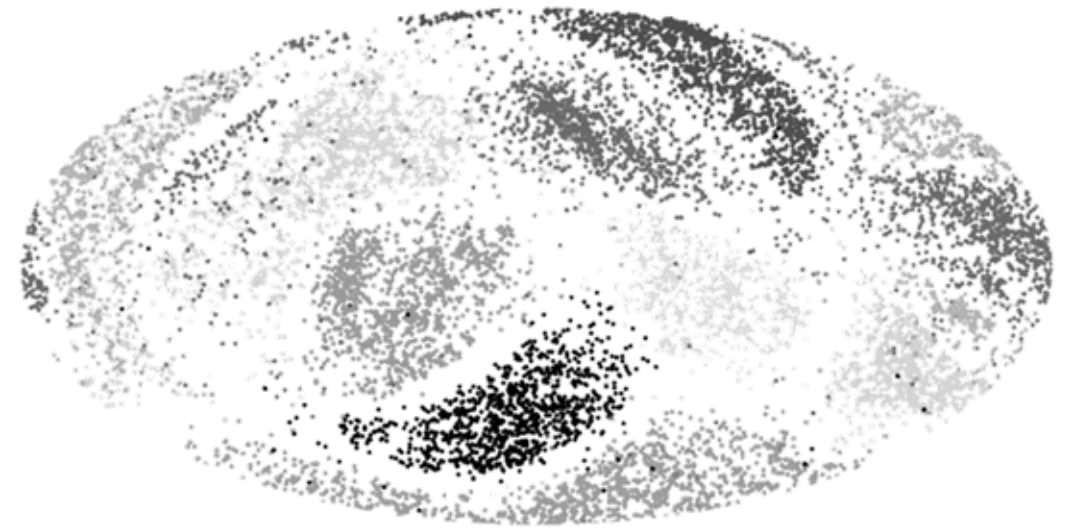
2022: Hyperspherical Variational Auto-Encoders, Davidson et al.

<https://arxiv.org/abs/1804.00891>

<https://github.com/nicola-decao/s-vae-pytorch>



(a) $\mathbb{R}^2$ latent space of the $\mathcal{N}$ -VAE.



(b) Hammer projection of S^2 latent space of the $\mathcal{S}$ -VAE.

0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9

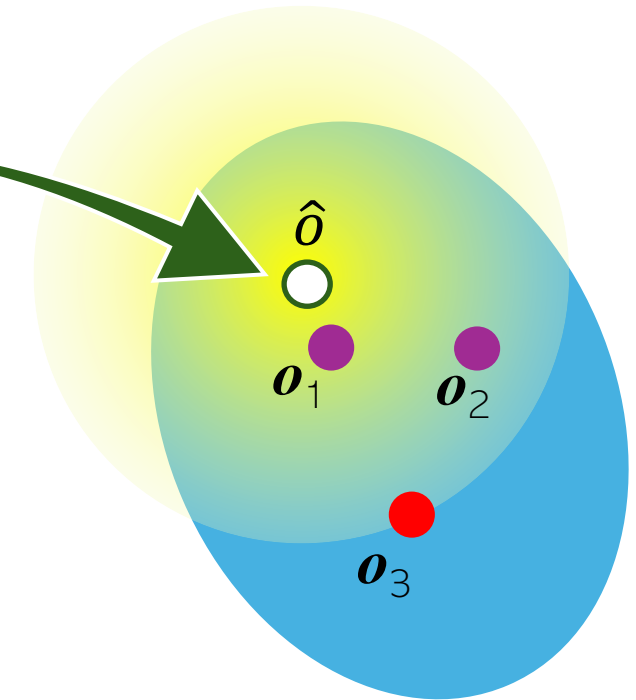
Figure 2: Latent space visualization of the 10 MNIST digits in 2 dimensions of both $\mathcal{N}$ -VAE (left) and $\mathcal{S}$ -VAE (right). (Best viewed in color)

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k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction

Person (Domain)	Animal (Domain)	Object (Domain)
Age ($\mathbb{N}$)	Height ($\mathbb{R}^+$)	Height ($\mathbb{R}^+$)
Height ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)
Gender ($\{M, F, N\}$)	Speed ($\mathbb{R}^+$)	Length ($\mathbb{R}^+$)
Occupation ($\{Student, Teacher, \dots\}$)	Type ($\{Mammal, \dots\}$)	Material ($\{Wood, Metal, \dots\}$)
Education ($\{BSc, MSc, \dots\}$)	Legs ($\mathbb{N}$)	Color ($\{White, Black, \dots\}$)
Experience ($\mathbb{N}$)	Lifespan ($\mathbb{N}$)	...
Location ($\{Canada, US, \dots\}$)	...	
Skill Level ($\{Novice, Experience, \}$)		
Income ($\mathbb{N}$)		
...		

 $e(o)$ 

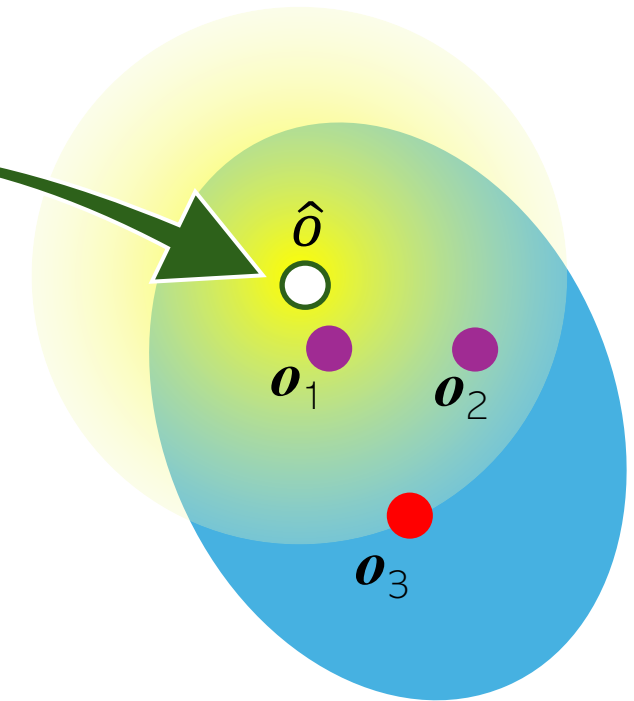
Heterogenous Space
Discrete Space

k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction → Feature Scaling

Person (Domain)	Animal (Domain)	Object (Domain)
Age ($\mathbb{R}^+$)	Height ($\mathbb{R}^+$)	Height ($\mathbb{R}^+$)
Height ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)	Weight ($\mathbb{R}^+$)
Gender ($\{0.0, 1.0, 2.0\}$)	Speed ($\mathbb{R}^+$)	Length ($\mathbb{R}^+$)
Occupation ($\{0.0, 1.0, \dots\}$)	Type ($\{0.0, 1.0, \dots\}$)	Material ($\{0.0, 1.0, \dots\}$)
Education ($\{0.0, 1.0, 2.0\}$)	Legs ($\mathbb{R}^+$)	Color ($\{0.0, 1.0, \dots\}$)
Experience ($\mathbb{R}^+$)	Lifespan ($\mathbb{R}^+$)	...
Location ($\{0.0, 1.0, 2.0\}$)	...	
Skill Level ($\{0.0, 1.0, 2.0\}$)		
Income ($\mathbb{R}^+$)		
...		

$e(o)$



Homogeneous Continuous Space

k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction $\rightarrow$ Feature Scaling

Age: $[1, 2, \dots, 100]$, Income: $[\$40,000, \dots, \$200,000]$

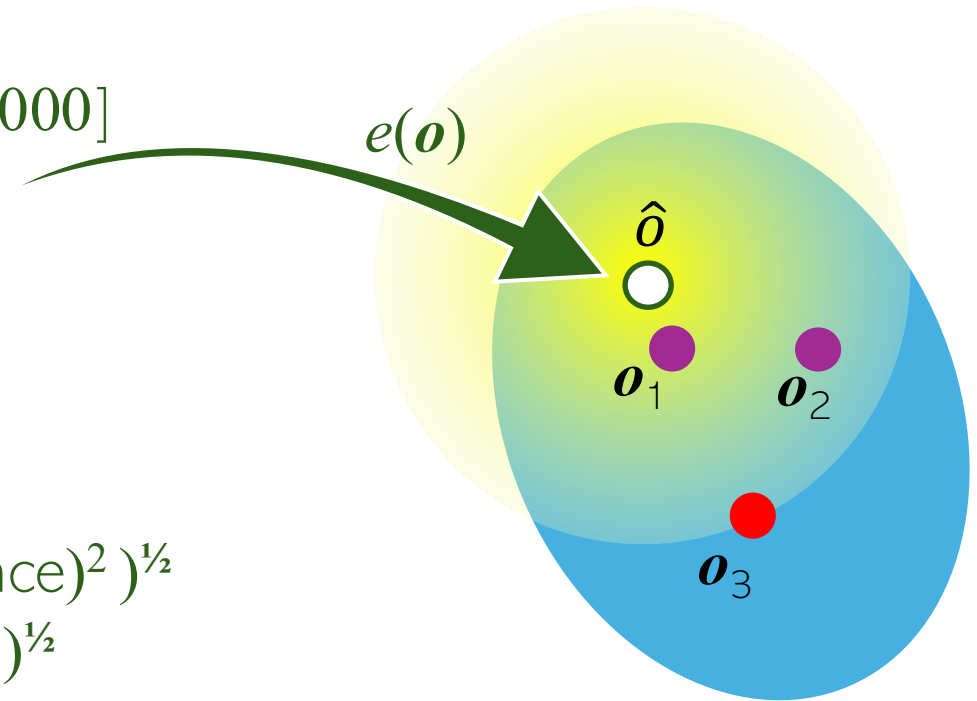
$e(\hat{o}: \text{Reia}) = (20, \$100,000)$

$e(o: \text{Hossein}) = (50, \$150,000)$

Euclidean Distance:

$$\begin{aligned} d(e(\hat{o}), e(o)) &= ((\text{age difference})^2 + (\text{income difference})^2)^{\frac{1}{2}} \\ &= ((20 - 50)^2 + (100,000 - 150,000)^2)^{\frac{1}{2}} \\ &= ((\mathbf{30})^2 + (\mathbf{50,000})^2)^{\frac{1}{2}} \end{aligned}$$

Income dimension dominates and overwhelms the age dimension.

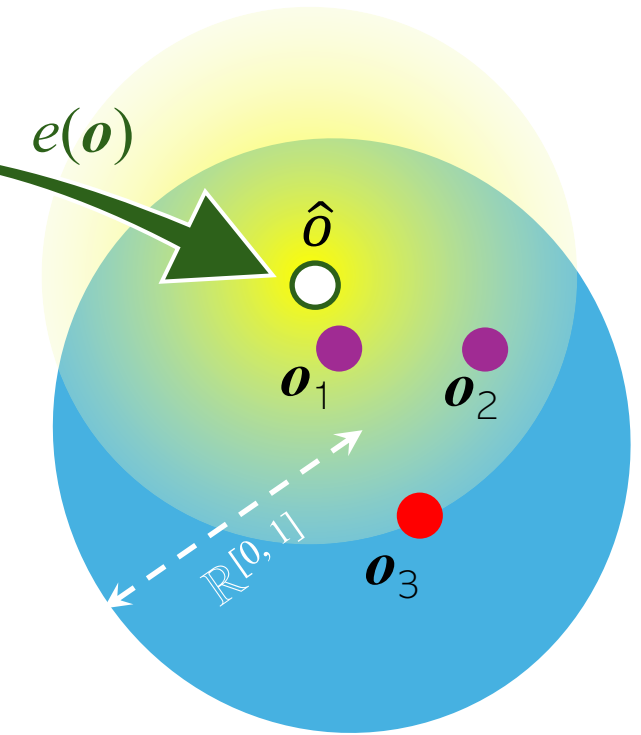


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k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction $\rightarrow$ Feature Scaling

Min-Max Normalization: $x' = \frac{x - \min}{\max - \min} \in \mathbb{R}^{[0, 1]}$



Unit Euclidian Space

Age: $[1, 2, \dots, 100]$, Income: $[\$40,000, \dots, \$200,000]$

$e(\hat{o}: \text{Reia}) = (20, \$100,000)$

$e(o: \text{Hosseini}) = (50, \$150,000)$

$\text{Min}_{\text{Age}} = 1, \text{Max}_{\text{Age}} = 100$, then Reia's age become $x' = \frac{(20 - 1)}{100} = 0.19$

$\text{Min}_{\text{Income}} = \$40,000, \text{Max}_{\text{Income}} = \$200,000$, then Reia's income become $x' = \frac{(\$100,000 - \$40,000)}{\$200,000 - \$40,000} = 0.375$

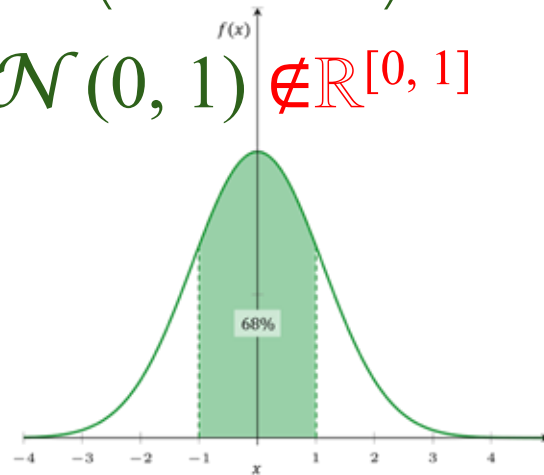
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k -Nearest Neighbor: Entities

Hands-Crafted Feature Extraction $\rightarrow$ Feature Scaling

Standard Normal (Gaussian) Distribution (z-score)

$$x' = \frac{(x - \mu)}{\sigma} \sim \mathcal{N}(0, 1) \notin \mathbb{R}^{[0, 1]}$$



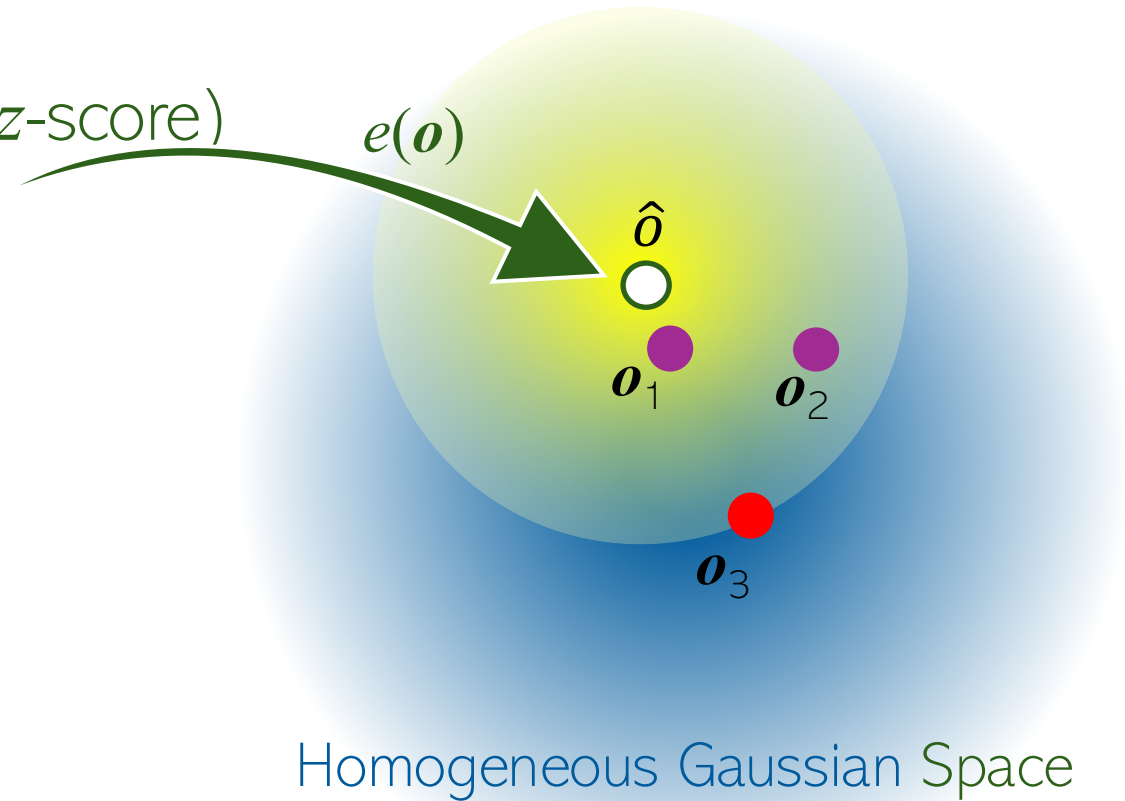
Age: [1, 2, ..., 100], Income: [\$40,000, ..., \$200,000]

$e(\hat{o}: \text{Reia}) = (20, \$100,000)$

$e(o: \text{Hossein}) = (50, \$150,000)$

$\mu_{\text{Age}} = 40, \sigma_{\text{Age}} = 10$, then Reia's age becomes $x' = \frac{(20 - 40)}{10} = -0.2$

$\mu_{\text{Income}} = \$80,000, \sigma_{\text{Income}} = \$20,000$, then Reia's income become $x' = \frac{(\$100,000 - \$80,000)}{\$20,000} = +1.0$



Homogeneous Gaussian Space

Outline: W05B

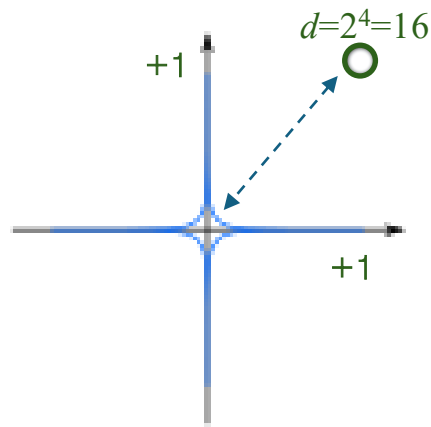
- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Final Notes and Food for Thoughts ...
- III) Evaluation

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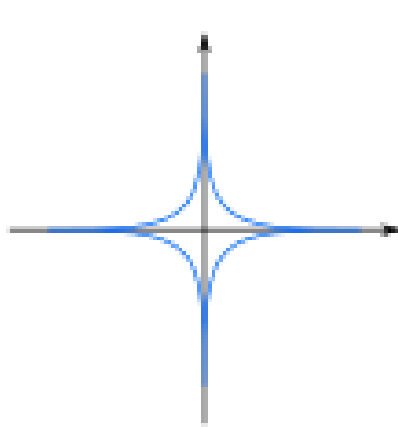
k-Nearest Neighbor: Minkowski's Other Distances

Blue lines show all points (x, y) with same distance 1.0 to the center $(0, 0)$

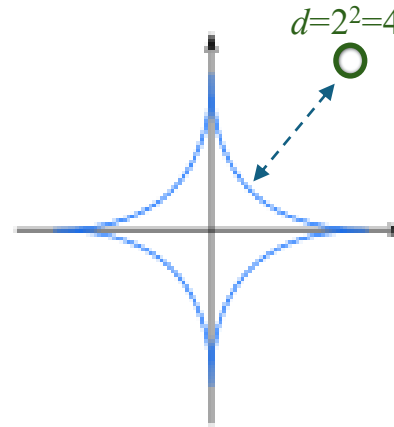
Look at the corner $(1, 1)$, how far is it? $d((1,1),(0,0))=(|1-0|^p+|1-0|^p)^{1/p}=(1^p+1^p)^{1/p}=2^{1/p}$, if $p \rightarrow \infty$, $2^{1/\infty}=2^0=1$



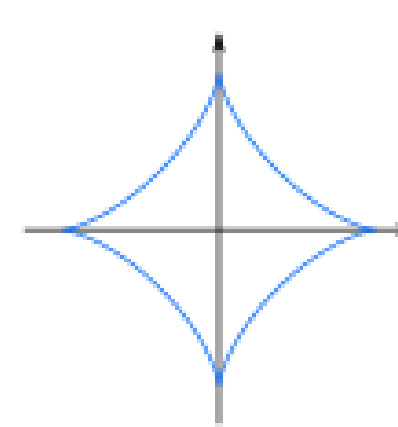
$$p = 2^{-2} \\ = 0.25$$



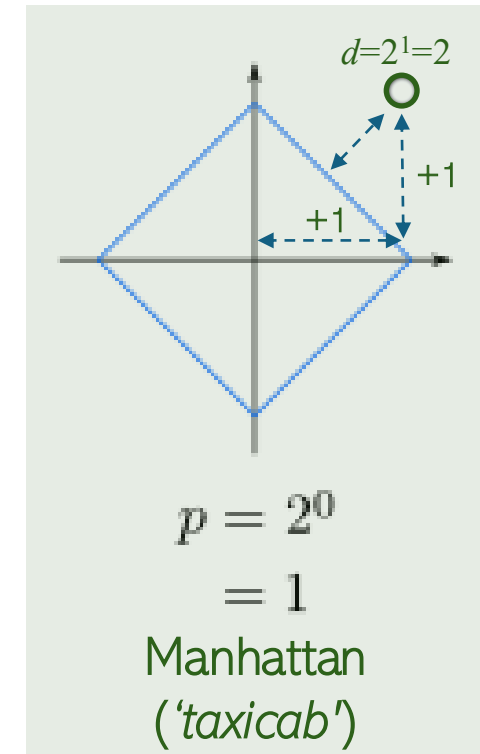
$$p = 2^{-1.5} \\ = 0.354$$



$$p = 2^{-1} \\ = 0.5$$



$$p = 2^{-0.5} \\ = 0.707$$



$$p = 2^0 \\ = 1 \\ \text{Manhattan} \\ (\text{'taxicab'})$$



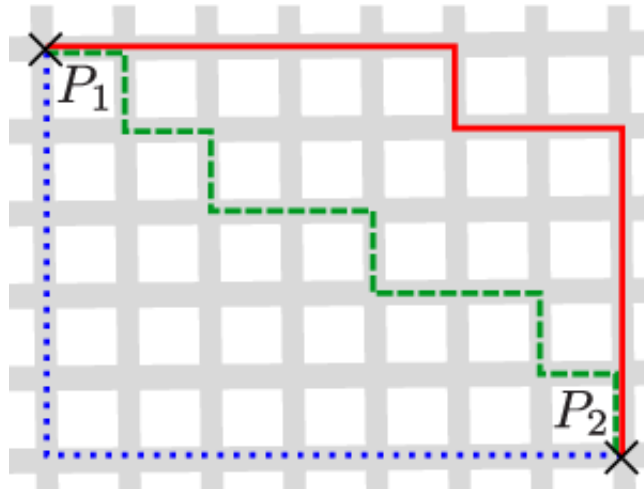
Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

Hermann Minkowski (1864 – 1909), Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen

Best known for his foundational work on space-time 4D space which facilitated geometric interpretations of Einstein's special theory of relativity (1905)

k -Nearest Neighbor: Manhattan Distance (Taxicab)



Mathematically,

- Sum of differences in components, or
- Sum of steps to take in each dimension.

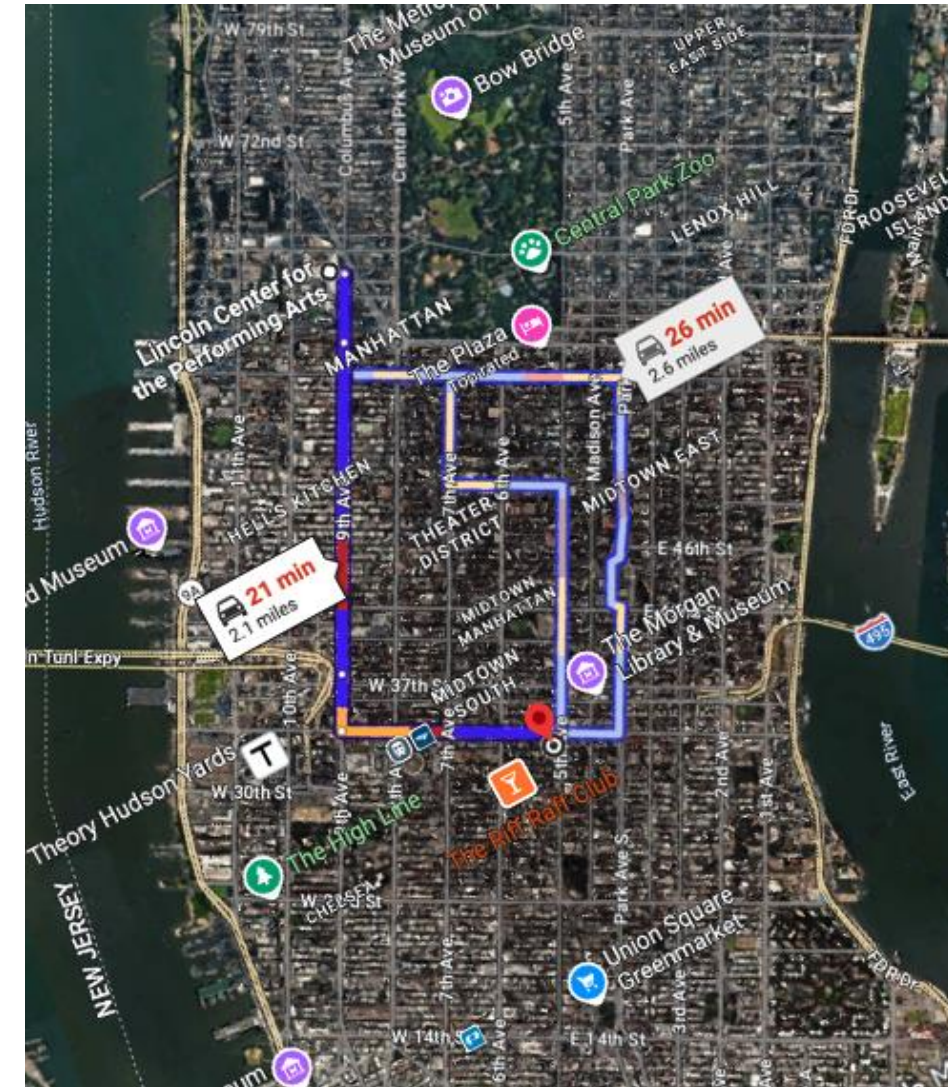


Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p} = \left(\sum_{i=1}^n |x_i - y_i| \right)$$

Hermann Minkowski (1864 – 1909), Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen

Best known for his foundational work on space-time 4D space which facilitated geometric interpretations of Einstein's special theory of relativity (1905)

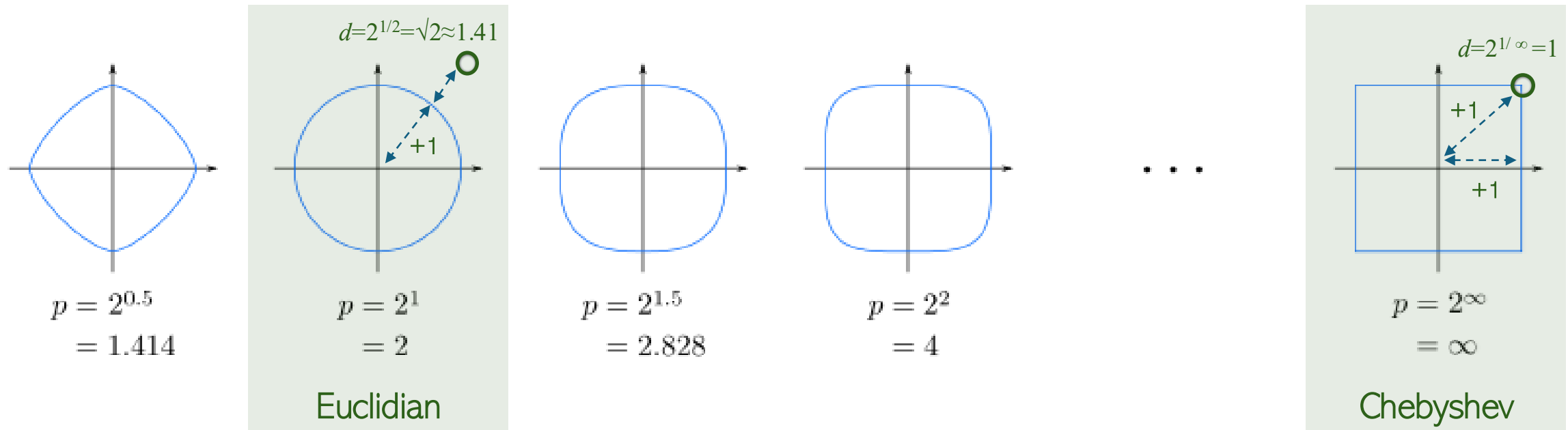


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k -Nearest Neighbor: Minkowski's Other Distances

Blue lines show all points (x, y) with same distance 1.0 to the center $(0, 0)$

Look at the corner $(1, 1)$, how far is it? $d((1,1),(0,0))=(|1-0|^p+|1-0|^p)^{1/p}=(1^p+1^p)^{1/p}=2^{1/p}$, if $p \rightarrow \infty$, $2^{1/\infty}=2^0=1$



Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

Hermann Minkowski (1864 – 1909), Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen

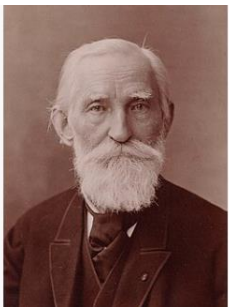
Best known for his foundational work on space-time 4D space which facilitated geometric interpretations of Einstein's special theory of relativity (1905)

k -Nearest Neighbor: Chebyshev Distances (Chessboard)

King's Distance: minimum number of moves for a king to reach a point. To a king, corner is as easily accessible as the right/left spots.

Mathematically,

- The largest differences among the components
- The largest step among the dimensions



$$\lim_{p \rightarrow \infty} \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p} = \max_i (|x_i - y_i|)$$

	a	b	c	d	e	f	g	h	
8	5	4	3	2	2	2	2	2	8
7	5	4	3	2	1	1	1	2	7
6	5	4	3	2	1		1	2	6
5	5	4	3	2	1	1	1	2	5
4	5	4	3	2	2	2	2	2	4
3	5	4	3	3	3	3	3	3	3
2	5	4	4	4	4	4	4	4	2
1	5	5	5	5	5	5	5	5	1
	a	b	c	d	e	f	g	h	

Pafnuty Lvovich Chebyshev (1821 – 1894), considered to be the father of Russian mathematics. Among his well-known students and Andrei Markov. According to the Mathematics Genealogy Project, Chebyshev has 17,533 mathematical descendants as of January 2025.

k -Nearest Neighbor: Minkowski's Other Distances

2	1	2
1		1
2	1	2

Taxicab

$\sqrt{2}$	1	$\sqrt{2}$
1		1
$\sqrt{2}$	1	$\sqrt{2}$

Euclidean

1	1	1
1		1
1	1	1

Chebyshev



Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

Hermann Minkowski (1864 – 1909), Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen

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k -Nearest Neighbor: Can be a multi-label classifier?

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1}[c_i = c] \right\}$$

Multi-Label Classification: Example
 Lazada Product Title Quality Challenge
https://github.com/hosseinfani/product_title_quality_cikm_2017



0 = (Titles in English)
C₁ Clarity = {1: Clear, 0: Not Clear}
C₂ Conciseness = {1: Concise, 0: Not Concise}

Clarity
 "Not very red clutch bag sack travel backpack unless cheap with free gift"

Conciseness
 "Not Sexy Teen Clever Women Wear Clever Look Cool Simple Style Casual Comfortable Messenger Bag Handbag Fashion Bag Tote Handbag Draw"

Why sellers do that?

$f(\text{"Not Sexy Red Clutch Bag Sack Travel Backpack ..."}) = (0: \text{Not Clear}, 0: \text{Not Concise})$
 $f(\text{"Bold Black Backpack with Striking Red Lips \& Teeth ..."}) = (1: \text{Clear}, 0: \text{Not Concise})$
 $f(\text{"Backpack"}) = (0: \text{Not Clear}, 1: \text{Concise})$
 $f(\text{"Black Backpack with Red Lips Graphic"}) = (1: \text{Clear}, 1: \text{Concise})$

k-Nearest Neighbor: Is it explainable (*x*AI)?

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1}[c_i = c] \right\}$$

Can it explain, justify, or provide evidence on why it picks a specific label?

To customers/clients:

- Decline a loan/mortgage/credit card/admission/claim application of a customer
- Fraud transaction/claim of a customer
- Diagnose a disease

To government/managers:

- Approve a loan/mortgage/credit card/admission/claim application of a customer
- Normal transaction/claim of a customer

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k -Nearest Neighbor: Pros & Cons

Pros:

Very simple and intuitive

Easy to implement

No explicit model assumptions like what would be voting strategy, distance function, ...

Naturally handles '*nonlinear*' decision boundaries (we cannot understand this now)

Useful baseline to compare with

Cons:

No training but at the cost of very expensive inference (linear to the training set size)

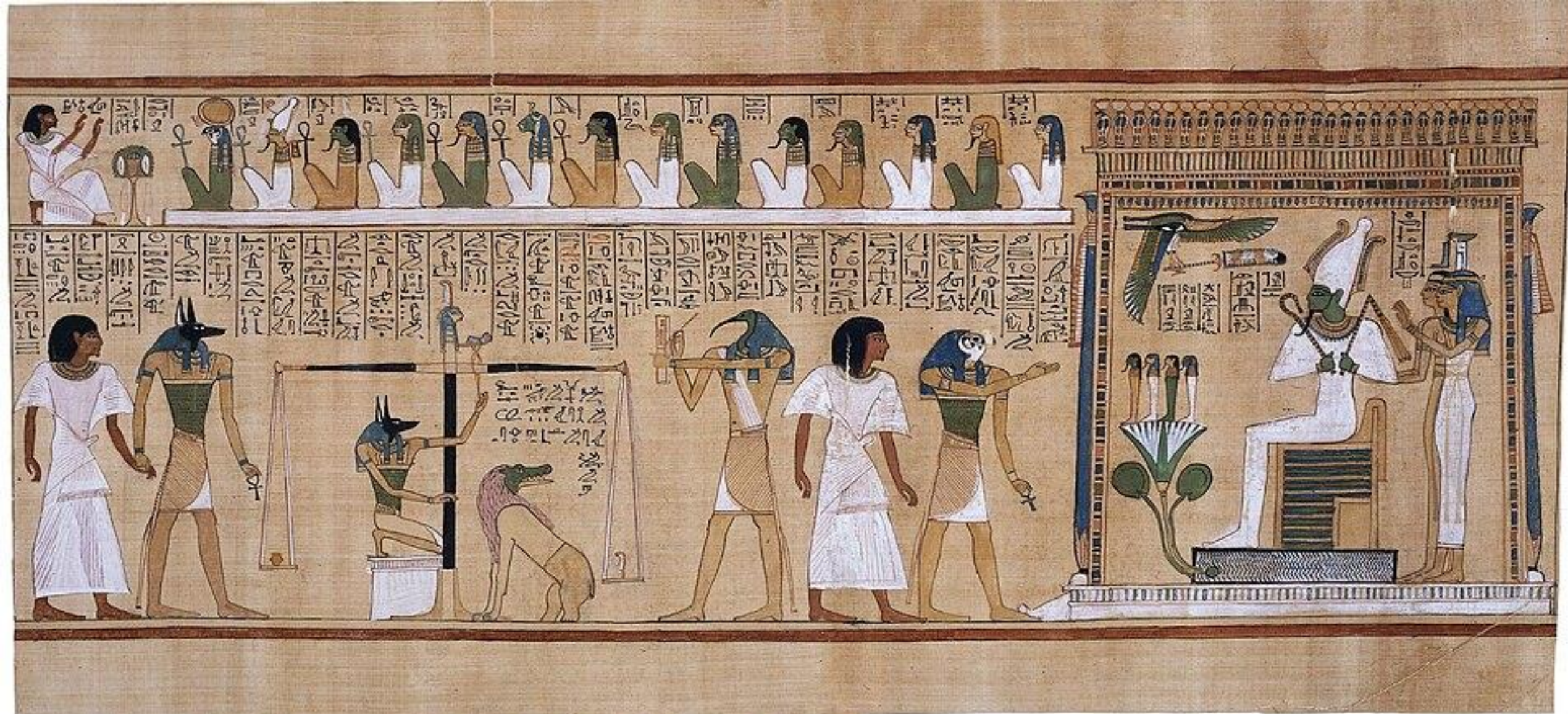
Requires storing training data

Sensitive to the choice of k , embedding space, distance function, feature scaling, ...

Poor in high dimensions (curse of dimensionality)

Outline: W05

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- IIII) Evaluation**
Metrics, Methodology



Duat /doo·et/ or Tuat /taat/ where people go after they die, in ancient Egyptian mythology. The '*Weighing of the Heart*' from the '*Book of the Dead*' from the Papyrus of Hunefer, dated to the 19th Dynasty around 1275 BCE. The deceased Hunefer is taken into the judgment hall by the deity Anubis, who weighs a portion of Hunefer's soul, represented by his heart. This ritual is completed with Ammit the Devourer awaiting the result, and Thoth recording. Next, the triumphant Hunefer, having passed the test, is presented by the falcon-headed Horus to Osiris, seated in his shrine with Isis, Nephthys and the four sons of Horus.

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Fully Supervised Classification Evaluation

Technique (k NN, LR, SVM, ...) does *not* matter ...

Students (technique) learn from the course (labeled set) differently, yet all are evaluated/judged the same.



Statue of Lady Justice blindfolded and holding a balance and a sword, outside the [Court of Final Appeal, Hong Kong](#)

$$(o, c) \rightarrow f(o) \cong c$$

Lady Justice, or the goddess Justitia, an allegorical personification of justice and moral force in ancient Roman art known as Iustitia or Justitia. Her attributes are scales, a sword and blindfold.

https://en.wikipedia.org/wiki/Lady_Justice

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Fully Supervised Classification Evaluation

Accuracy

$$(o, c) \rightarrow f(o) \cong c$$

$$(q, a) \rightarrow f(q) \cong a$$

For many (o, c) , how often $f(o) = c$...

For many (q, a) in the exam, how often $Hossein(q) = a$...

For a random (o, c) , what's the probability prior to answer: $\mathcal{P}(f(o) = c \mid (o, c))$

For a random (q, a) , what's the probability (before exam) $Hossein(q) = a$? $\mathcal{P}(Hossein(q)=c \mid (o, c))$

Fully Supervised Boolean Classification Evaluation

Accuracy

$$(o, 1) \rightarrow f(o) \cong 1$$

$$(o, 0) \rightarrow f(o) \cong 0$$

For many (o, c) , how often $f(o) = 1$ if $c = 1$, and how often $f(o) = 0$ if $c = 0$...

Being in a binary choice (True/False) exam ...

For a random (o, c) , what's the probability

A random question from the pool of binary choice questions

$$\mathcal{P}(f(o) = c \mid (o, c)) = \mathcal{P}(f(o) = 1) \times \mathcal{P}(c = 1) + \mathcal{P}(f(o) = 0) \times \mathcal{P}(c = 0)$$

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Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** (fair, equal) distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

For 50 instances of $(o, 1)$, how often $f(o) = 1 \dots$

For 50 instances of $(o, 0)$, how often $f(o) = 0 \dots$

For a random (o, c) , what's the probability

$$\mathcal{P}(f(o) = c \mid (o, c)) = \mathcal{P}(f(o) = 1) \times \mathcal{P}(c = 1) + \mathcal{P}(f(o) = 0) \times \mathcal{P}(c = 0)$$

$$= \mathcal{P}(f(o) = 1) \times \frac{50}{100} + \mathcal{P}(f(o) = 0) \times \frac{50}{100}$$

$$= [\mathcal{P}(f(o) = 1) + \mathcal{P}(f(o) = 0)] \times \frac{1}{2} \text{ ("average on accuracies of two labels")}$$

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Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

A **biased pessimist** $f(o) = 1$ unless proved otherwise! What's accuracy?

A **biased optimist** $f(o) = 0$ unless proved otherwise. What's accuracy?

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Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

A biased pessimist $f(o) = 1$ unless proved otherwise! 50 out 100 hit: $\frac{50}{100} = 0.5$

A biased optimist $f(o) = 0$ unless proved otherwise. 50 out 100 hit: $\frac{50}{100} = 0.5$

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Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

An **unbiased drunken** $f(o) = 1$ or 0 randomly, who cares! What's accuracy?

A drunk student during exam, what would be the exam score?

An **unbiased gambler** $f(o) = 1$ or 0 by flip/toss a coin ... What's accuracy?

A gambler student during exam, what would be the exam score?

Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!

An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...

$$\begin{aligned} \mathcal{P}(f(o) = c) &= \mathcal{P}(f(o) = 1) \times \mathcal{P}(c = 1) + \mathcal{P}(f(o) = 0) \times \mathcal{P}(c = 0) \\ &= \mathcal{P}(f(o) = 1) \times \frac{50}{100} + \mathcal{P}(f(o) = 0) \times \frac{50}{100} \end{aligned}$$

w/ equal probabilities to guess 1 or 0, so $\frac{1}{2}$ each:

$$= \frac{1}{2} \times \frac{50}{100} + \frac{1}{2} \times \frac{50}{100} = 0.5$$

Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

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	Accuracy
A biased pessimist $f(o) = 1$ unless proved otherwise!	0.5
A biased optimist $f(o) = 0$ unless proved otherwise.	0.5
An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!	0.5
An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...	0.5

"If only the universe were fairly distributed and balanced... if only ..."

Fully Supervised Boolean Classification Evaluation

Accuracy: **balanced** distribution of classes

$$(o, 1)_{001 \dots 050} \rightarrow f(o) \cong 1$$

$$(o, 0)_{051 \dots 100} \rightarrow f(o) \cong 0$$

	Accuracy
A biased pessimist $f(o) = 1$ unless proved otherwise!	0.5
A biased optimist $f(o) = 0$ unless proved otherwise.	0.5
An unbiased drunken $f(o) = 1$ or 0 randomly, who cares!	0.5
An unbiased gambler $f(o) = 1$ or 0 by flip/toss a coin ...	0.5
A data mining technique (method)	0.50000001 at least
The Oracle of Delphi	1.0



Taxi Driver, 1976, Martin Scorsese.